

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	T Jyothi Reddy	Reg. No.:	192472079
Section / Batch:	2024	Date:	

Code of Conduct

I T Jyothi Reddy(192472079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T Jyothi Reddy

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

- 1. Maintain entity coherence using assignment, mistakes, and you.**
- 2. Use a Cause–Effect or Elaboration discourse relation.**
- 3. Include at least two keywords from:**
 - **review**
 - **improve**
 - **confident**
 - **feedback**
- 4. Response length must be 2–3 sentences.**

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	4	4	3	3	18	90
2	4	4	4	3	3	18	
3	3	3	4	4	4	18	

Faculty In-charge	Course Coordinator	Program Director
Dr. T. Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Reference Resolution and Discourse Coherence

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	Book
they	Ravi and Arun
the book	Book

b) Constraint-Based Resolution

Referring Expression	Valid Antecedent	Reason
He	Ravi	Gender and number agree, and Ravi is the main subject of the previous sentence.
him	Ravi	Arun helped Ravi find the book, which is semantically meaningful.
it	Book	“Find it” refers to the book Ravi was looking for.
they	Ravi and Arun	“They” is plural and refers to both people.
the book	Book	It refers to the previously mentioned book.

The constraints used are gender and number agreement, recency, grammatical role, semantic compatibility, and discourse coherence.

c) Final Coreference Chains

Ravi → He → him

Arun → Arun

Book → it → the book

Ravi + Arun → they

d) Rewritten Paragraph

Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

e) Ambiguity

If the sentence “He was looking for a book” appeared before Arun was introduced, the pronoun “He” would refer to Ravi if Ravi had already been introduced. If no suitable male person had been introduced, the pronoun would have no clear antecedent, causing

ambiguity and reducing discourse coherence.

Q2. Dialog Act Recognition and Response Generation

a) Required Dialog Acts

1. Reassure
2. Advise

b) Important Entities

1. Assignment
2. Mistakes
3. You

c) Three Possible Responses

Response 1:

You have already submitted the assignment, so try not to worry too much. You can review the feedback later and use it to improve your work and feel more confident next time.

Response 2:

It is normal to feel worried after submitting an assignment, but mistakes can help you improve. Review the feedback carefully so you can learn from it and become more confident.

Response 3:

Do not worry too much; you have completed and submitted your assignment. Use the feedback to review any mistakes and improve your future work.

d) Comparison of Responses

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
Response 1	High	High	High	High
Response 2	High	High	High	High
Response 3	High	High	High	High

e) Best Response and Justification

Response 2 is the best response because it clearly reassures the student and provides useful advice. It maintains entity coherence, uses the required keywords, follows a cause-effect relationship, and maintains a positive and polite tone.

f) Effect of Poor Entity Coherence

If entity coherence is not maintained, the response may become confusing, irrelevant, and difficult to understand. The connection between the assignment, mistakes, and the student may be lost.

Q3. Word Sense Disambiguation and Meaning Representation

a) Possible Meanings of “Bank”

1. Bank as a financial institution.
2. Bank as the side of a river.
- 3.

b) Correct Sense and Justification

The correct sense of “bank” is the side of a river.

The words “boats,” “water,” and “moving across” provide contextual evidence. Priya can sit beside a river and watch boats moving across the water. Therefore, the financial institution meaning is not compatible with the given context.

c) Predicate Logic Representation

Let:

Priya(p) = Priya is a person

Bank(b) = b is a river bank

Boat(z) = z is a boat

Water(w) = w is water

The predicate logic representation is:

$\text{Priya}(p) \wedge \text{sit}(p,b) \wedge \text{bank}(b) \wedge \text{beside}(p,b) \wedge \text{watch}(p,z) \wedge \text{boat}(z) \wedge \text{move}(z) \wedge \text{water}(w)$

d) Unambiguous Paraphrase

Priya sat beside the river and watched the boats moving across the water.

e) How WSD Improves Machine Translation

Word Sense Disambiguation identifies the correct meaning of a word based on its context. For example, “bank” may mean a financial institution or the side of a river.

Identifying the correct sense helps the translation system select the appropriate meaning and reduces translation errors. Therefore, WSD improves semantic accuracy, contextual correctness, and coherence in Machine Translation.